EXP NO:19 Experiment to understand Logistic Regression for a given data set.

AIM:
To understand the Logistic Regression algorithm for the data set given.

PROGRAM:

python

```python
import numpy as np
import pandas as pd

# Read the dataset
df = pd.read_csv("/content/student_admission.csv")
df

# Display first 5 rows
df.head()

# Check missing values
print(df.isnull().sum())

# Rows with no label can't be used for training, so drop them
df = df.dropna(subset=['Admitted'])

# Fill missing feature values
df['Age'] = df['Age'].fillna(round(df['Age'].median()))
df['EntranceScore'] = df['EntranceScore'].fillna(round(df['EntranceScore'].mean()))

# Reset index
df.reset_index(drop=True, inplace=True)

# Confirm no missing values remain
print(df[['Age', 'EntranceScore', 'Admitted']].isnull().sum())

# Select features and label
features = df.iloc[:, [1, 2]].values
label = df.iloc[:, 3].values

# Import libraries
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

# Try different random states
for i in range(1, 401):
    x_train, x_test, y_train, y_test = train_test_split(
        features, label, test_size=0.2, random_state=i
    )

    model = LogisticRegression(max_iter=1000)
    model.fit(x_train, y_train)

    train_score = model.score(x_train, y_train)
    test_score = model.score(x_test, y_test)

    if test_score > train_score:
        print("Test {} Train {} Random State {}".format(
            test_score, train_score, i
        ))

# Split dataset for the final model
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=42
)

# Train final Logistic Regression model
finalModel = LogisticRegression(max_iter=1000)
finalModel.fit(x_train, y_train)

# Display training and testing accuracy
print(finalModel.score(x_train, y_train))
print(finalModel.score(x_test, y_test))

# Classification report
from sklearn.metrics import classification_report

print(classification_report(
    label,
    finalModel.predict(features),
    zero_division=0
))
```

OUTPUT:

```
Department       1

Age              1

EntranceScore    1
```

```
Admitted          0
dtype: int64
Age               0
EntranceScore     0
Admitted          0
dtype: int64
Test 1.0 Train 0.75 Random State 1
Test 1.0 Train 0.75 Random State 3
Test 1.0 Train 0.75 Random State 5
Test 1.0 Train 0.75 Random State 13
Test 1.0 Train 0.75 Random State 14
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0.0
              precision    recall  f1-score   support
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          No       0.67      0.67      0.67         3

         Yes       0.86      0.86      0.86         7


    accuracy                           0.80        10

   macro avg       0.76      0.76      0.76        10

weighted avg       0.80      0.80      0.80        10
```

RESULT:

Thus the program to understand the Logistic Regression algorithm for the given data set was executed successfully.